

Hong-Phuc Lai

GenAI Researcher – Deep Learning Engineer

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RESEARCH INTERESTS

My research focuses on **Efficient and Robust Multimodal Artificial Intelligence**, advancing performance by reducing training and inference costs while ensuring trustworthy behavior. My expertise spans **Diffusion Models**, **Flow Matching**, and **Reinforcement Learning** with a current focus on applying these algorithms to Embodied AI by designing **Vision Language Action (VLA)** models for complex perception and control in humanoid robotics.

EDUCATION

Hanoi University of Science and Technology University

B.Sc. in Automation Control Engineering; GPA: 3.64/4.00

Hanoi, Vietnam

Aug 2018 – Aug 2022

- **Talent Program:** Selected among top 30 of 500 students
- **Selected Coursework:** Linear Algebra (A), Probability & Statistics (A), Linear Control Theory (A+, Highest Score), Introduction to Robotics (A), Computer Vision (A+, Highest Score)

TECHNICAL SKILLS

Languages: Python, C/C++, MATLAB

Technologies: Pytorch, OpenCV, Accelerators, Transformers, Diffusers, Git, Latex

Research Areas: 3D Grid Representations, Novel View Synthesis, Text-to-3D Generation, Point Cloud Segmentation, Efficient Diffusion Models, Autoregressive Video Diffusion, Reinforcement Learning, Deep Learning

EXPERIENCE

Predoctoral Research Resident

VinAI Research - Qualcomm AI Research

Aug 2024 – Present

Hanoi, Vietnam

- Supervisor: Principal Scientist [Dr. Anh Tran](#).
- Proposed a multi-scale extension of 3D grid representations, demonstrating superior performance in novel view synthesis, video fitting, and text-to-3D generation.
- Developed a novel, efficient inference-time diffusion algorithm capable of generating high-resolution 4K images in under 20 seconds, achieving state-of-the-art quality and up to 30× faster speed.
- Introduced the first successful reinforcement learning post-training algorithm for few-step Flow models, addressing a previously unsolved problem.
- Implemented a full pipeline for panoptic segmentation of large-scale 3D point cloud scenes, deployed in Qualcomm XR systems; achieved segmentation of 90 million points in under 2 hours.

Applied Scientist

VinBrain - NVIDIA

Oct 2023 – Aug 2024

Hanoi, Vietnam

- Implemented an improved Grad-CAM algorithm for medical image classification, providing clearer and more interpretable explanations of model predictions.

Computer Vision Engineer

Viettel Aerospace Institute

Aug 2022 - Aug 2023

Hanoi, Vietnam

- Developed and deployed efficient real-time computer vision models for image classification and object detection, leveraging knowledge distillation, advanced augmentation, and multi-GPU training to achieve high accuracy and improved mAP in production systems.
- Kickstarted a real-time face anti-spoofing project using anomaly detection and depth auxiliary signals, achieving high accuracy and laying the foundation for future deployment.

PUBLICATIONS

[PixelRush: Ultra-Fast, Training-Free High-Resolution Image Generation via One-step Diffusion](#)

Hong-Phuc Lai, Phong Nguyen, Anh Tran

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026) - **Highlight Presentation**

[Paspp medical transformer for medical image segmentation](#)

Hong-Phuc Lai, Thi-Thao Tran, Van-Truong Pham

International Conference on Data Science and Applications (ICDSA 2023) - **Best Paper Award**

[Axial attention mlp-mixer: A new architecture for image segmentation](#)

Hong-Phuc Lai, Thi-Thao Tran, Van-Truong Pham

IEEE International Conference on Communications and Electronics (ICCE 2022)

PREPRINTS

[PixelFlash: Eliminating the VAE Bottleneck for Ultra-Fast High-Resolution Image Generation](#)

Hong-Phuc Lai, Phong Nguyen, Anh Tran

Under review

AWARDS & ACHIEVEMENTS

Highlight Presentation

2026

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

Recognized among the top 3% of all accepted papers

Best Paper Award

2023

International Conference on Data Science and Applications (ICDSA)

Awarded for outstanding research contribution and best paper presentation

Best Presentation Award

2022

School of Electrical Engineering, HUST

Honored for excellence in thesis presentation among graduating students

Gold Medal in Mathematics

2016

Thai Binh Specialized High School

First place in mathematics entrance exam for gifted high schools across the Northern Delta (Grade 10)

OPEN SOURCE PROJECTS

Projects demonstrate both coding quality and research capability, implementing and reproducing algorithms from published papers.

Diffusion-deep-image-prior | [GitHub](#)

- This project implements two novel optimization-based image inpainting approaches using diffusion models: DIIP (Diffusion Image Prior, *NeurIPS 2025*), which leverages pre-trained diffusion models as image priors, and SDI (Self-Diffusion for Solving Inverse Problems, *ICCV 2025*), which trains a U-Net from scratch for image restoration.

Kernel Density Steering | [GitHub](#)

- This project reproduces "Kernel Density Steering: Inference-Time Scaling via Mode Seeking for Image Restoration" - a Google img2img method that applies mode-seeking strategies to enhance diffusion-based image restoration.

HiWave | [GitHub](#)

- This project provides a minimal reimplementation of the core algorithm from "HiWave: Training-Free High-Resolution Image Generation via Wavelet-Based Diffusion Sampling," created purely for educational purposes.